# Chapter 15
# Equity and Participation in Decisions: What Can Nanotechnology Learn from Biotechnology in Kenya?

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*Discussions of equity and technology often focus on the distribution of risks and benefit. But key to addressing these issues is how power to make decisions that shape those issues is distributed. In the last chapter, Bal reported that members of the American public expressed a desire for equity once they were given a voice in nanotechnology decision making. In this chapter, Matthew Harsh looks at how biotechnology regulation was developed in Kenya to help think about how nanotechnology might be regulated in a developing country. A variety of horizontal inequalities criss-cross his account, but they are different from the categories presented in the first part of this book: Europeans versus Kenyans; government versus non-government; and experts versus lay people. The agora is not open; existing power constellations are built into access to meetings and where decisions are made. Harsh shows that the ultimate distribution of risk is certainly not influenced most by those who are affected most, namely, those who live in rural areas. The lesson is a sobering one with regard to both the complexity and obdurateness of politics. Harsh argues that if equity concerning the effects of nanotechnology is to be achieved, equity of access to decision making about nanotechnology will have to come first.—eds.*

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## 15.1 Introduction

When examining news about the latest developments in nanotechnology, it appears that nanotechnology offers ever-increasing promise for developing countries. For example, nanotechnology is being used to develop a coating for sandy soil so that the soil will retain more water and be more fertile (Baldwin 2009). Nanotechnology could provide better water filtration (Dickson 2009), enable easier HIV testing (Gill 2009), and produce more options for generation of energy (Bourzac 2009). These technologies, it is said, will benefit (poor) people living in developing countries. However, as this volume shows, even if nanotechnologies such as these become technically feasible, it is unlikely that the benefits of new nanotechnologies will be distributed evenly. Distributional consequences of new nanotechnologies raise questions about equity and equality: Who benefits more from these technologies and who less? Who is susceptible to more risk or less risk? Who *should* benefit more?

While distributional outcomes of nanotechnologies are undoubtedly important, the intent of this essay is to examine equity and nanotechnology in a converse way (see Fig. 15.1). I focus on unequal inputs to the nanotechnology development process. By inputs, I am not referring to the material or labor inputs used in technology creation, although there are certainly equity issues surrounding these kinds of inputs (see chapters in Part I of this volume). Rather, I am referring to *decision inputs* that guide and shape the nanotechnology development process (see the top half of Fig. 15.1). Conceived broadly, these are decisions about which technologies to develop and how. As opposed to questions about who benefits and who is at risk, the equity and equality questions here are: Who decides? Who should decide? For instance in the case of the nano-coating for soils mentioned above, relevant questions include: How much have farmers who might use the coating in their arid fields taken part in decisions about the development of the coating? Will they take part in decisions about how it might be deployed in their region or regulated by their national government? How much should they be involved in all these decisions? How does that compare with how much scientists, environmental organizations, or manufacturers take part in these same decisions?

The goal of this essay is not to give direct answers to these types of questions. Rather, this essay looks laterally across questions about participation in decision making from a perspective of equity. The essay asks: How does focusing on equity help us understand these important questions about "who decides"? Or more explicitly: How is equity relevant to analysis of participation in decision making about nanotechnology in developing countries or regions?[1]

I must state a few caveats about context and comparison before going forward. Context is utterly important—the answers to questions about who decides and who should decide will be different depending on the specific nanotechnology and the specific country or region in which it is to be developed. Context is especially important for this essay because I use a comparative method. I draw out lessons for nanotechnology by reflecting on recent developments in agricultural biotechnology in Kenya. The nanotechnology-biotechnology comparison has been made many times before (see for instance Mehta 2004; David and Thompson 2008)

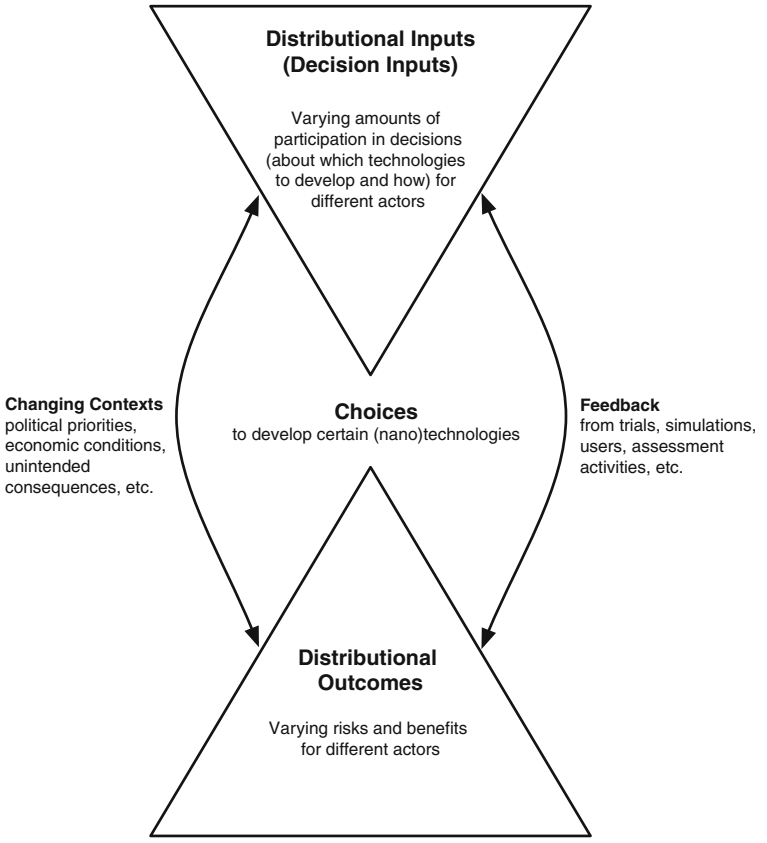



**Fig. 15.1** A distributional view of technology development

and has its shortfalls (Mody 2007). Comparisons usually focus on how nanotechnology and biotechnology are technically similar and different and how probable impacts of the technologies vary in terms of risks and benefits. The overall goal of such comparisons is often to try to ensure that public reactions to nanotechnology are less adverse than previous reactions to biotechnology, particularly to biotechnologies involving genetic-modification (GM). Making these comparisons between nanotechnology and biotechnology is fraught with difficulty, not least because of the broadness and still emergent nature of both nanotechnology and biotechnology. Furthermore, previous nanotechnology-biotechnology comparisons have also not focused on developing countries or regions. In light of these criticisms, this essay avoids technical comparisons and comparisons of claimed impacts. It instead focuses on *process* and decision making that leads to technology development. The choice to focus on decision making is partly due to the fact that it is still too early to judge many of the impacts of agricultural biotechnologies in Kenya, particularly GM technologies, because no GM crops have been commercialized in Kenya at the time of writing.

The essay presents two arguments about how equity is relevant to analysis of participation in decisions about nanotechnology in developing regions. The first is that equity is relevant because there are clear inequalities surrounding access to decisions about emerging technologies. Experience from biotechnology in Kenya shows that who can and cannot access decisions depends on the vertical dimension of inequality (inequalities between the rich and poor) and the horizontal dimension (inequalities between groups distinguished by culturally-defined differences such as race, gender, and religion) (Cozzens and Wetmore 2008). Horizontally, the Kenya case shows how science itself becomes a basis for culturally-defined difference that shapes the distribution of access to decisions. Furthermore, focusing on equity helps address criticisms of other approaches to analyzing participation in decision making, which can lose sight of differential agency resulting from varying access to resources or from social and cultural hierarchies. The lesson for nanotechnology is that, like all new technologies, nanotechnologies will be developed among existing inequalities that affect access to decisions. However, especially to the extent that nanotechnology utilizes more complex and advanced science, it could create a new basis for unequal access to decisions based on unequal access to new scientific knowledge.

While uneven access to any given decision is clearly important, the possibilities for how, and if, certain decisions can even occur are also strongly influenced by institutional constraints. My second argument is that focusing on equity can help us understand how institutional mechanisms[2] structure decisions and thus influence access and participation. Different institutional mechanisms lead to different distributions of when and where decisions occur. Compared to the linear institutional mechanisms of the Green Revolution model of agricultural research and development, new partnership models at work in Kenya have offered some actors more access to decisions. However, this increased access could be considered to be just or unjust depending on the theory of distributive justice adopted. The lesson here for nanotechnology is that it is important to look beyond the more headline-catching technological innovations and focus on the perhaps more mundane realm of changing institutional structures behind nanotechnology developments, because these will shape participation in decisions. Any new institutional mechanisms must in turn be examined in historical context. Previous institutional trends have led to uneven development of biotechnology in Africa, and the same could be true for nanotechnology.

In the conclusion I reflect on how focusing on unequal inputs helps provide a more complete picture of equity. It is also a step towards understanding how governance can be more anticipatory (Barben et al. 2008) and technology assessment might be more real-time (Guston and Sarewitz 2002) in development contexts.

## 15.2 Distribution of Access

Ideas about *access* are the key to my first argument about how equity is relevant to analyses of participation in decisions about emerging technologies in developing regions. Discussions of equity generally begin by examining something that people

value (for example, something tangible such as food or intangible such as education) and a population or society among which this valued or desired thing is unevenly distributed (Cozzens et al. 2008). Access to decisions about emerging technologies is such a thing. Experience from agricultural biotechnology in Kenya shows how access to decisions is both highly valued and highly unevenly distributed.[3]

The development of agricultural biotechnology involving GM in Kenya over the last two decades[4] embodies the trajectory of an emerging technology, especially as related to uncertainty. In the early emergence period of GM in Kenya (in the 1990s), high levels of technical uncertainty existed about whether certain technologies could actually be developed. For instance in 1991, Kenya began a project to create a GM virus-resistant sweet potato. The project involved a partnership between the Kenya Agricultural Research Institute (KARI), the United States Agency for International Development, the private company Monsanto, and later a non-governmental organization (NGO) called the International Service for the Acquisition and Application of Agricultural Biotechnologies (ISAAA) (ABSP 2002).[5] Much has been written about this project in the academic literature and the international media, as it was the first attempt to develop and cultivate a GM crop in East Africa, and in all of sub-Saharan Africa except South Africa (New Scientist 2004; Odame et al. 2003a, b; Qaim 1999). The initial research and development was performed in the United States, and in 2000 GM plants were transferred to Kenya for confined testing. By 2003 tests showed that the sweet potato plants failed to be resistant to Kenyan viruses because they were engineered to be resistant to American strains of the viruses (Cohen and Paarlberg 2004; Gathura 2004 Interview with former KARI scientist, 2005). As late as 2003 then, there was still uncertainty about the technical viability of the GM sweet potato.

Technical uncertainty surrounding GM in Kenya has decreased since 2004, yet other intimately connected non-technical uncertainties have not decreased as quickly. The sweet potato project has been regrouping its technical efforts, attempting to create a new cultivar that hopefully fares better against Kenyan viruses (Interview with manager of NGO, 2004). Several other GM crop development projects are underway in Kenya including insect resistant cotton and maize and virus-resistant cassava (See Harsh 2005). These projects have been bolstered by commercialization of similar crops in other countries (including some African countries).[6] Despite these technical efforts, there is still a great deal of uncertainty about how GM crops will reshape Kenyan farming systems and broader Kenyan society and political economy. Projection analyses of GM crops have been performed but these mainly use quantitative socioeconomic methods to justify the development of GM technologies, such as that of Qaim for the sweet potato project (1999). Missing in these analyses are discussion of cultural and social uncertainties that are harder to quantify and will likely decrease at a much slower pace than technical uncertainties.[7]

It is the possible pervasive and far reaching effects of GM technologies, coupled with uncertainty, that make decisions about GM so important and make access to those decisions such a highly sought after commodity. This can be seen in examining some of the primary and most visible decision points about the development of GM

in Kenya. For the most part, these decision points are related to the establishment of Kenya's national biosafety system.

Biosafety involves regulation to protect human and natural environments from hazards associated with the release of a new species. Biosafety is one of the five "policy venues" that, according to Robert Paarlberg, constitute the broader decision making or governance of biotechnology. The other four are: "intellectual property rights (IPR) policy; trade policy; food safety and consumer choice policy; and public research policy" (2000, 4).

National governance of biotechnology in Kenya essentially defaulted to decisions about biosafety mainly because capacity to govern GM, like the development of GM technologies themselves, was mostly donor-driven, and biosafety was the specific focus of many donor efforts. One of the most significant donor-sponsored biosafety projects was coordinated by the United Nations Environment Program-Global Environmental Facility (UNEP-GEF) (UNEP-GEF 2003).[8] Eventually biosafety became operationalized in the form of Kenya's Biosafety Bill and joint Biotechnology and Biosafety Policy.[9] A key issue for governance of biotechnology in Kenya thus becomes examining how the Biosafety Bill and joint Policy were created. Specifically for my argument about the distribution of access, it is important to ask: Who was and was not involved in making decisions about the Bill and Policy?

The Biosafety Bill and Policy have had long and bumpy histories[10] that have provided a hodgepodge of opportunities for participation of different kinds of actors. Workshops were a main mechanism for participation, and thus main decision inputs. Each donor initiative sponsored several workshops that shaped decisions about the Bill and Policy. The workshops were often not solely organized by the donor or the coordinating government body, the National Biosafety Committee (NBC).[11] They were often co-organized or solely organized by NGOs, such as ISAAA (the NGO that led the GM sweet potato project mentioned above), the Agricultural Biotechnology Stakeholders Forum (ABSF), the Kenya Agricultural Biotechnology Platform, and later Africa Harvest Biotech Foundation International (AHarvest). As organizers, these NGOs had privileged access to biotechnology decisions, and also the ability to effectively exclude other organizations or individuals by not informing them about workshops.[12] These NGOs had further access in that they were the force keeping the development of biosafety moving forward between donor initiatives. This meant sometimes using their own funding to sponsor their own workshops (interview with senior policy maker in Ministry of Education, Science and Technology, 2005). Eventually, government-mandated comment periods provided the most formal mechanism for participation of broader actors in the shaping of the Biosafety Bill. Finally, less formal ways of participating in shaping the Biosafety Bill and joint Policy chosen by some actors included carrying out protests, distributing press releases, and writing editorials and letters to the editor in local newspapers.

Mid 2004–2005 was a particularly active and turbulent time for biotechnology and biosafety in Kenya, and one that highlights disparity of access to decisions about biotechnology. In early 2004, a coalition of NGOs formed whose goal was

to stop the development of GM crops in Kenya (hereafter referred to as the anti-GM coalition). The coalition included at least five NGOs. Two were national branches of large well-known international NGOs (the headquarters of one was in Europe and the other in Africa) that work on poverty alleviation and a number of general issues related to rural poor livelihoods. Another NGO was a smaller European-based organization which worked only on GM issues. The coalition also included an NGO that was a regional (i.e., sub-national) network of community-based organizations that worked with small-scale farmers on land management. One of these community-based organizations was a small-scale farmers association and was itself an active member of the anti-GM coalition. All of these NGOs came to work on GM through their advocacy work related to the environment, small-scale farming, or ecological land management. The groups engaged in biosafety work in addition to their other projects. Thus, they were not well-funded when it came to work on biotechnology or biosafety. Of all the NGOs, only one person was employed to work full-time on biotechnology issues. The staff members from these groups that participated in the anti-GM coalition were generally Kenyan (only one was European) but were not scientists.

The coalition never gained much access to decisions about biotechnology and biosafety, a situation that—in the eyes of the coalition—led to bad policy choices, and certainly led to a non-constructive policy environment. When the coalition first formed, it began to investigate how the biosafety process had occurred to date, and tried to participate in the process. The coalition wrote letters to the NBC and requested meetings. However, they did not receive responses to their letters. They were denied entry to meetings and denied copies of the draft Bill and Policy. None of the NGOs ever heard about any workshops being held (interviews with coalition members, 2004). A general lack of communication between the coalition and the NBC further fueled the coalition's use of ideological and conspiratorial language when discussing the actions of the NBC and KARI in press releases and public meetings. This led to antagonistic relationships between the NBC and the coalition. It also led to the adoption of highly polarized viewpoints which prevented opportunities for any real debate about how biotechnology should be developed in Kenya (Harsh 2005, 2008). Overall, the coalition charged that they were left out of decisions about biotechnology and biosafety, and that because of this, the interests of small-scale farmers and the environment were left out of the NBC and of decisions about biotechnology (See Harsh 2009). This coalition clearly desired access to decisions, but had very little success in attaining it.

On the other hand, in this same period ISAAA and ABSF had a relative abundance of access to decisions about the development of biotechnology and biosafety. As opposed to the NGOs that make up the anti-GM coalition, ISAAA and ABSF are very well-funded, and working on biotechnology and biosafety development constitutes their core missions. Senior staff members of ISAAA and ABSF are generally Kenyan and hold graduate degrees in the sciences. During 2004–2005, key members of these organizations were active members of the UNEP task forces whose job was to draft the Policy and Bill. Also during this period and part of the UNEP program, ISAAA and ABSF organized several workshops. The crucial role that ABSF

played in developing biosafety was noted by a top official in the National Council for Science and Technology (Interview, 2005). One NGO employee claimed that ABSF effectively wrote the Biosafety Bill and had the ear and personal mobile phone numbers of key policy-makers (Interview, 2008).

Both vertical and horizontal dimensions of inequality (Cozzens and Wetmore 2008) are clearly relevant to this story about unequal access to decisions about biotechnology in Kenya. With respect to the vertical dimension (inequalities between the rich and poor), the Kenyan case shows that access to resources is clearly an important factor that influences access to decisions. However, it is not just that well-funded NGOs and actors often have more access to decisions than less well-funded actors. The coordinating body in the Kenyan Government, the NBC, also claimed not to have enough resources to coordinate with all the NGOs interested in biosafety and biotechnology decisions. Moreover, perhaps the poorest group of actors in the story, small-scale farmers, had very little access to decisions. Generally, only the very small percentage of farmers that worked with NGOs on either side of the biotechnology issue knew anything about the development of biotechnology or biosafety in Kenya. And even of this small subset, only a smaller subset could come to Nairobi for workshops and meetings, depending on whether NGOs could pay for their travel and expenses.

The horizontal dimension of inequality (inequalities between groups distinguished by culturally-defined differences such as race, gender and religion) is also clearly important. In the case of decisions about agricultural biotechnology in Kenya, race and gender are significant, but possibly not the most significant factors. Perhaps paradoxically, the main actors within key NGOs who have had the most access to decisions (such as ISAAA, ABSF, and AHarvest) were women and African (native Kenyans).[13] In the period of late 2004–2005 the only NGO staff member who was European was a member of the anti-GM coalition—the side with significantly *less* access.

The most significant culturally-defined difference that led to inequalities in the Kenyan case was science itself. The staff members of NGOs in the anti-GM coalition were not scientists. Senior staff members of ISAAA, ABSF and AHarvest were generally PhD scientists. Members of the anti-GM coalition repeatedly stated that it was primarily this difference that kept them out of decisions about biosafety and biotechnology. Because they were not scientists and lacked scientific expertise, they were not able to participate in decisions about biotechnology (Interviews, 2004 and 2005). As in Europe and North America, decisions about GM in Kenya were framed in a scientific manner (Levidow 2007; Rayner 2003); the focus was on risks to human health or environment that could be quantified by scientific-based risks assessment. If actors were not fluent in scientific parlance, or could not frame their concerns in a risk-based scientific manner, they had a very difficult time accessing and participating in decisions (Harsh 2009).

Examining this case from Kenya shows that equity is relevant to participation in decisions about emerging technologies in developing regions because access is a highly desired and unevenly distributed commodity. Equity is also relevant though, because focusing on equity addresses criticisms of other frameworks that analyze participation in decisions. An equity analysis that examines horizontal and vertical

inequalities, as in the Kenyan case study, brings differential agency of actors to the forefront.

Other approaches to participation in decision making do not focus so explicitly on how horizontal and vertical inequalities keep certain actors from decisions. For instance, participation in policy decisions is a major topic of discussions in political science and international relations discussions about governance (Hajer and Wagenaar 2003; Rhodes 1997).[14] Discussion of governance in this context focuses on collectivity in decision making, but does not necessarily problematize how varying access to resources or social ties can prevent certain actors from getting a seat at the table (Levidow 2007; Pestre 2008; Harsh and Smith 2007). In development studies since the 1970s there has been a focus on participation of the poor in their own development (e.g. Chambers 1983) that has led to many participatory policies, methodologies, and practices in development agencies. However, activities done in the name of participation can maintain or reinforce social hierarchies, such as those within a village (Cornwall and Pratt 2003). Participation and governance are discourses that can serve to cover up the politics of development initiatives; they depoliticize development (Ferguson 1994). An equity approach to participation can perhaps bring politics to the surface with its focus on inequalities. Some evidence for this comes from literature in Science and Technology Studies that has been focusing on the politics of participation involving science for some time (see for example Petersen 1984, and the other chapters in this section of this volume). Equity and justice are issues that are explicitly discussed (Nelkin 1984; Krimsky 1984).

There are clear lessons for nanotechnology from this first argument about access. The most obvious lesson is that no matter what nanotechnologies are being developed, or in which developing region they are being developed, they will be developed among existing inequalities that will affect access to decisions. These inequality concerns will not be as straightforward as who has access to resources and who does not, but access to resources will always be one key issue. Importantly, nanotechnologies could create the basis for new horizontal inequalities based on science in the same way that biotechnology has done in Kenya. Especially to the extent that nanotechnology will draw on more advanced or complex science, the distribution of who has access to this scientific knowledge will shape the distribution of who has access to decisions about how technologies should be developed, regulated, and used.

## 15.3 Institutional Structuring

My second argument is that equity is relevant to understanding participation in decisions about emerging technologies in developing regions because focusing on inequality issues can help show how different institutional mechanisms structure decision points, and thus who can access decisions and participate in them. By institutional mechanisms I refer to the configuration of public and private organizations (at international, national, and sub-national scales) that fund and carry out

research and development of emerging technologies specifically for use in developing regions. Different institutional mechanisms can lead to different spatial and temporal distributions of decision points for different actors. In terms of *space* one can ask: In what geographical area do decisions occur, in what cities and countries? In what settings do they occur? Do they occur in laboratories, board rooms, or government offices? In terms of *time* one can ask: When in the technology development process does a given actor have opportunities to participate in decisions? Is it in upstream activities such as deciding which research projects to fund? Or is it in more downstream activities such as decisions about regulation, distribution, or use? I return to agricultural biotechnology in Kenya to give examples of how institutional configurations structure distributions of decisions.

The Green Revolution has had a significant impact on institutional mechanisms for developing agricultural technologies in Kenya, and throughout other developing regions, since the 1960s. The Green Revolution set out to increase food production by providing high-yielding crop varieties to farmers in Asia, Africa, and South America (Pearse 1980). Based on the model of land grant universities in the United States, scientific knowledge and new crop varieties, along with new fertilizers, pesticides, and irrigation technologies, were produced in international research laboratories as technological packages. These were then passed to national agricultural research centers (such as KARI in Kenya) and then to farmers via state-run extension services (Clark et al. 2003). Figure 15.2 illustrates the model, often called the "pipeline model" (Clark et al. 2003, 1847). The whole process was largely donor supported (some of the most prominent donors were the Ford and Rockefeller Foundations).

This model is still influential in agricultural technology development largely because it has created entrenched international and national institutional structures. The international research centers established by the Green Revolution

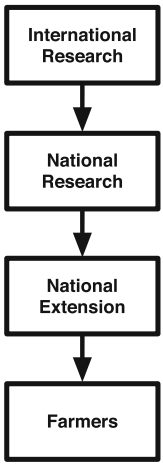



**Fig. 15.2** Institutional model of the Green Revolution

are collectively known as the Consultative Group for International Agricultural Research system (CGIAR 2009). There are now 15 CGIAR centers world-wide, two of which have their headquarters in Kenya. Nationally, Kenya's agricultural research system is one of the most developed in all of Africa. Over twenty-five institutes conduct agricultural research in Kenya and 833 full-time equivalent employees were employed in the late 1990s (Beintema et al. 2003). Kenya spends over three times as much as the average sub-Saharan African country on agricultural research (Beintema et al. 2003).

The Green Revolution model significantly structures the distribution of decision points about what technologies should be developed. Spatially, there are two important issues. Initial decisions about what crops to develop happen in laboratories and test fields at research institutes. They also often happen far from the country and region of eventual use. Temporally, under the Green Revolution institutional model, major decisions that lock in the specific technologies to be developed occur upstream, early on in the development process. As knowledge moves linearly down the pipe (see Fig. 15.2) and technologies are developed, there are fewer options to control technology development; there are fewer decision points open to actors. Generally, the only decision in which farmers (end users) can participate is whether or not to adopt the new technologies and techniques. For the initial high yielding varieties of the Green Revolution, biosafety was not an issue as it is for GM, so few discussions or debates took place like those described above related to regulating GM in Kenya.

Since at least the 1990s, several new institutional mechanisms to develop agricultural technologies have emerged in Kenya.[15] These models essentially involve reconfiguring institutions; they are attempts to form new kinds of connections or new partnerships between institutions other than the linear connections associated with the Green Revolution model. These new mechanisms also involve new actors.

For instance, a now common model is the development of new technologies via Public-Private Partnerships (PPPs). The motivations for moving to these types of partnerships forge together technical, legal, economic, and moral concerns. The logic is that for new advanced biotechnologies, it is the private sector, and not the public sector, which possesses the relevant scientific expertise and owns the relevant intellectual property necessary to develop technologies. The private sector thus partners with a public sector research institute (a national research institute such as KARI or one of the international CGIAR research institutes) by sharing knowledge with the public partner and by licensing or donating intellectual property. The public partner gains access to new technologies, and the private partner benefits from a more philanthropic public image as well as the development of possible new markets (Horsch and Montgomery 2004). Due to the convergence of multifaceted motivations, donors are widely supporting these kinds of partnerships. Some of these are the same donors that funded the Green Revolution (such as the Rockefeller Foundation), but some major new donors have also emerged, such as the Bill and Melinda Gates Foundation. The GM sweet potato project has been developed via this kind of PPP mechanism, as are many of the other current GM projects in Kenya.

In addition to the private sector, NGOs are another important category of new actors prevalent in new institutional mechanisms. NGOs are often co-ordinating actors in PPPs, liaising between public and private actors, as well as with local farmers, policy-makers, and local university scientists. ISAAA is playing this role for the sweet potato project. NGOs can also play more specialized roles, such as the African Agricultural Technology Foundation (AATF, also based in Nairobi) whose role is specifically to negotiate intellectual property rights in PPPs.

Beyond PPPs, other new types of partnership arrangements are also emerging in Kenya. One such mechanism is direct collaboration between national research institutes (or other similar institutions) in two or more countries (so-called South-South partnerships or North-South partnerships depending on the countries involved). Another mechanism is direct collaboration between a national and an international agricultural research center to co-develop a specific product in a specific context, such as animal vaccines (Smith 2005).[16] Lastly, an important new model is the development of Centers of Excellence which bring public sector expertise from several countries together, focused on one area of science, such as the Bioscience for Eastern and Central Africa initiative which was founded in 2004 and is based in Nairobi (ILRI 2003).

Compared to the Green Revolution model, these new partnership mechanisms seem inherently more systemic. They implicitly or explicitly encourage more interconnections between different kinds of actors. However, in attempting to forge new connections between actors, how have partnership models changed when and where decision points occur and thus who can be involved in decisions?

The evidence from Kenya suggests that the distribution of decisions has not changed as much as one might expect given the greater focus in PPPs on partnership and forming new linkages, and on broader views of technological change. Yet, the changes in distribution of decisions that have occurred have enabled some new actors to have more access to decisions compared to the Green Revolution model.

For instance in the GM sweet potato project, ISAAA argues that despite the failed trials the project has actually been a success (Interviews, 2004). According to ISAAA, if one adopts a broad view of technological change, then even though a technically viable product has not emerged, the project was a success because it helped develop capacity to regulate GM technologies in Kenya that did not exist before the project began.[17] However this more systemic view did not significantly alter where and when decisions occur. The story of the genesis of the project involves a Kenyan scientist doing her doctoral training in the United States (Odame et al. 2003a). Similar to the Green Revolution model then, initial decisions that locked in the specific technological choice were still mainly made outside Kenya and occurred upstream. One scientist explicitly stated that farmers did not need to be involved in the sweet potato project until a technical product was fully created (Interview, 2005).

There is further evidence that other PPPs involving GM in Kenya have operated with similar distributions of decisions as that of the sweet potato project. When asked about how initial decisions were made to begin GM projects in

Kenya, one actor[18] told the story of traveling to Australia for a scientific conference with a KARI scientist. After learning about the current state of GM science, the two Kenyan scientists were inspired to try to create GM products in Kenya (Interview, 2004). Regarding the specific decision to use GM to develop insect resistant maize in Kenya, a university scientist/NGO manager told the story of sitting around a board room table at the International Maize and Wheat Improvement Centre (a CGIAR center). A gentleman from the private company Novartis happened to be at this meeting and it was decided that work should begin on GM maize (Interview, 2004). That work evolved into a PPP now called the Insect Resistant Maize for Africa initiative. Spatially, these vignettes show how decisions were made in settings such as board rooms at research facilities and at scientific conferences. Some key decisions were clearly made geographically far from Kenya. Temporally, the decisions again locked in technological choices very upstream.

Nevertheless, new partnership mechanisms have changed the distribution of decisions in some significant ways. One main way is that NGOs, and to some extent private sector actors, now have much more access to decisions and more control of technology developments because of the move to PPPs. ISAAA is just one of many NGOs that play a managing role in PPPs in Kenya and throughout Africa.[19] In 2005, AHarvest received US$16.9 million from the Gates Foundation to coordinate a partnership to develop bio-fortified sorghum (AHarvest 2005). AATF received US$47 million from the Gates Foundation and the Warren G. Buffet Foundation to develop maize that is more water efficient. For many NGO staff members, working at NGOs gives them more access to decisions about biotechnology than they might have had in previous and alternate careers as university researchers. To a much lesser extent, the shift to PPP mechanisms also has increased the amount of access to decisions for some farmers. NGOs, such as ISAAA and AHarvest, generally have networks of farmers with which they work. Oftentimes these farmers are geographically close to NGO offices. Essentially, the situation is that some rural poor farmers (i.e., one group considered to be disadvantaged in an equity analysis) are gaining some additional access to decisions through these new institutional mechanisms.[20] But that increase pales in comparison to the increased access that NGO staff members (i.e., one group considered to be advantaged) are gaining because of new partnership mechanisms.

Given these changes, does the shift from the Green Revolution model to more partnership mechanisms represent a just intervention? This depends on the theory of distributive justice adopted.[21] From a libertarian perspective, the answer is most likely yes. Justice in this perspective focuses on preservation of property rights, and at the heart of PPP mechanisms are licensing agreements and contracts to protect intellectual property. From a utilitarian perspective that focuses on growth and not distributional issues (such as the distribution of access to decisions) the answer is also probably yes. The motivation for GM in African policy discourses is often linked to economic growth (Harsh 2009). From a contractarian perspective, the answer is again, yes. The disadvantaged actors are slightly better off because of PPP mechanisms. Some rural farmers are gaining more access to decisions. Finally, from

a communitarian perspective where just interventions decrease the gap between advantaged and disadvantaged actors the answer is surely no. The gap in access to decisions between the advantaged and disadvantaged actors is not decreasing, and probably even increasing. Scientists working for NGOs are experiencing a much larger increase in access than poor farmers.

There are evident lessons for nanotechnology from this discussion about institutional mechanisms and biotechnology. Given that institutional constraints strongly structure the distribution of decisions, it is important to pay as much attention to changes in institutional structures supporting the development of nanotechnology in developing regions as to changes in technical developments. Institutional developments rarely make headlines the way technical stories can. However, studying emerging institutional developments will reveal if the PPP model will be the dominant model for nanotechnology development, or if new models will emerge. Some new developments are already occurring. For example, the Gates Foundation recently announced a partnership with the United States National Science Foundation (NSF). The Gates Foundation will give US$24 million to the NSF (NSF will match this with an additional US$24 million) for research relevant to developing regions (including nanotechnology) and allow the NSF to vet projects (Doughton 2009).

A final point is that any new institutional mechanisms that do emerge to develop nanotechnology need to be historically contextualized. A lesson from the Green Revolution is that institutional strategies lead to infrastructural entrenchment that can create geographic disparities in development of new technologies. Partly because the Green Revolution led to significant infrastructural investments in Kenya, later biotechnology investments were also made in Kenya (favorable biosafety policies also played a role). The same cycle may be repeating itself again. PPP mechanisms are creating new infrastructure in terms of new networks of NGOs, many of which are based in Kenya. This is an important lesson for nanotechnology, especially to the extent that nanotechnology development will be infrastructure-dependent.

## 15.4 Conclusion

Through comparison with agricultural biotechnology in Kenya, this essay has argued that equity is relevant to analyses of participation in decisions about nanotechnology in developing regions. A focus on equity helps us understand that uneven distribution of access to decisions is linked to vertical inequalities related to access to resources, as well as to horizontal inequalities related to scientific identity and access to scientific knowledge. Framing analyses in terms of equity brings differential agency and politics to the forefront. Furthermore, distributional analyses can help us see how institutional constraints structure where, when, and even if, decisions take place.

There are perhaps some larger benefits of focusing on equity in analysis of participation in decision making. Such a focus provides a more complete and complementary approach to the study of equity and nanotechnology (see Fig. 15.1). This is not merely about providing analytical symmetry, balancing a focus on distributional consequences or outputs with a focus on unequal inputs, but about providing a means to think about how the two are connected. Inputs and consequences are not connected in a linear way. Deliberate feedback mechanisms are often put in place to provide information about possible outcomes based on technical trials, simulations, or modeling (see Fig. 15.1). Changing contexts, such as shifting political priorities, changing economic conditions, or unintended consequences, can all also have bearing on technological trajectories and their outcomes. In the end however, decision making usually involves speculation about distributional consequences and performing some sort or cost/benefit or risk/benefit calculation. This not only involves weighing the benefits that some will receive against the risks that others will face—a difficult political calculus made even more difficult because of uncertainty and unforeseen consequences—but also involves determining what constitutes a risk and a benefit. These calculations and constitutions could both drastically change depending on who is taking part in a decision. The distribution of access to decisions is thus importantly linked to distributional consequences.

In making my arguments to this point, I have avoided normative arguments about which different actors should have access to which decisions. Likewise, I have not argued in favor of one institutional mechanism to develop a nanotechnology over another. However, decision makers regularly need to make these commitments.

The second larger benefit of focusing on equity and participation in decisions, then, is that it helps equip decision makers with the ability to make normative commitments. For instance, one way that has been suggested to deal with uncertainty and unintended consequences of technology, is to make governance more anticipatory (Barben et al. 2008) and assessment of technologies more real-time (Guston and Sarewitz 2002). The conceptual framework of anticipatory governance involves creating anticipatory knowledge about how technologies might develop and how these developments are related to societal outcomes (Barben et al. 2008). The methodology of real-time technology assessment (RTTA) has tried (in developed countries) to implement anticipation by engaging with publics and end users of technologies and feeding their views into anticipatory knowledge. Anticipatory governance and RTTA thus generally try to widen the distribution of decision inputs and analyze and develop new feedback mechanisms between decisions and outcomes. In the context of institutional mechanisms to develop emerging technologies in developing regions discussed in this essay, implementation of an RTTA approach is most consistent with a communitarian perspective of distributive justice. From this perspective, one would argue for more access for farmers to decisions earlier on in the technology development process, and for institutional mechanisms that can do a better job of applying this so-called bottom-up approach than PPP mechanisms have been able

to do thus far for GM crop projects in Kenya. Just what these institutional mechanisms might look like, and indeed how governance might generally be made to be more anticipatory in developing regions, requires much more research. However, a starting point might be exploring how donors and other architects of emerging technological interventions in developing regions think about societal outcomes in their current institutional models.

## Notes

1. I use the term developing regions instead of developing countries to account for variations within countries.
2. By institutional mechanisms I mean the configurations of institutions (various public and private organizations operating at international, national, or sub-national scales) that fund and carry out research and development of technologies specifically for developing regions.
3. Information in this section and the next comes from ethnographic fieldwork carried out in Kenya from 2004 to 2008. Facts coming from individual interviews are cited as such.
4. Kenya has been engaging with other non-GM biotechnologies such as biopesticides and tissue culture since the 1960s, see the timeline in Harsh (2005).
5. Other partners that have more recently joined the project include the Roodeplaat Vegetable and Ornamental Plant Institute of South Africa, another public sector institute, and the Danforth Plant Science Center in the USA (Horsch and Montgomery 2004).
6. As mentioned above, currently there are no GM crops commercialized in Kenya. However, three other countries in Africa have commercialized GM crops. These are South Africa, Egypt and Burkina Faso.
7. For instance, in another (non-GM) agricultural biotechnology project involving tissue culture bananas developed by ISAAA (the same NGO that is a partner in the GM sweet potato project), farmers were concerned about how tissue culture bananas might lead to the disappearance of a banana variety important for betrothal ceremonies (Harsh and Smith 2007).
8. Bilateral support to develop biosafety in Kenya also came from the Netherlands Directorate-General for International Co-operation and the United States Agency for International Development. To be sure, some donor initiatives did lead to the development of the other policy venues that make up the governance of biotechnology, but these were far outshone by efforts to develop biosafety. See Harsh (2005 and 2008) for more detail.
9. The Biosafety Bill essentially provides the legal authority to regulate biotechnology. The joint Biotechnology and Biosafety Policy is a statement of intent to develop biotechnology in Kenya.
10. The first draft of the Biosafety Bill was produced in 2003 and the Bill was finally passed into law in 2008. Some of the factors leading to the prolonged history include: the signing and ratification of the Cartagena protocol in 2000 and 2002 respectively; and the decision in 2002 to create a new law to provide legal authority for regulating biotechnology instead of using existing laws, which was the strategy up to 2002. Drafting of the Biotechnology and Biosafety Policy started in 2002–2003 and the Policy was accepted by the Cabinet in 2006. (Policies in Kenya do not need Parliamentary approval.) See Harsh (2005 and 2008) for more details.
11. At the time the NBC was within the National Council of Science and Technology, a division of the Ministry of Education Science and Technology. With the passing of the Biosafety Bill into law in 2009, the NBC became the National Biosafety Authority. The hosting ministry was also split after the 2007 elections and is now the Ministry of Science and Technology.

12. For some workshops invitees were limited, for instance to only Members of Parliament. However, most workshops were nominally open to the public, but were not advertised broadly. Thus if one was not invited, one would generally not know about the workshop and would not be able to attend.
13 I have previously argued that these actors represent what Tvedt (2006) calls a new elite class of actors development system (Harsh 2009).
14. Here, governance is used to describe a change in the nature of decision making of states from one where the state operated in a command-and-control manner to one where the state is but one actor negotiating amongst other actors to make policy decisions, such as NGOs and private sector organizations (Pierre and Peters 2000).
15. These new institutional mechanisms have not entirely displaced pipeline mechanisms. Experience varies between different research centers and different developing regions.
16. This is different than the Green Revolution model in the that the international and national centers are jointly researching and developing the same product. This involves two-way knowledge flow and partnership. For instance, the international center must focus much more on specific national and even sub-national contexts.
17. Importantly, this was a revision from ISAAA's prior statements about the motivation for the project that focused more narrowly on the technological product—on the economic impact that the GM potato would have.
18. This actor was highly involved in the Nairobi biotechnology community having worked as a university scientist, a manager of an NGO, a coordinator for a donor project on biosafety, and as the head of a new research partnership at various overlapping points in her career.
19. Note that although many of the PPPs are based in Kenya, their remit is to develop GM technologies for all of Africa.
20. For example, one such group of farmers working with ISAAA is the Murang'a Banana Marketing Collective, a network created by ISAAA located in Murang'a district of Central Province approximately 70 km from Nairobi. Through their interactions with ISAAA they have learned about tissue culture technology. However, their access to decision making related to the technology is mostly limited to a decision to adopt the technology or not. In general, ISAAA does not release figures about how many farmers they worked with on the tissue culture project. See Harsh and Smith (2007) and Smith (2007).
21. The four theories of distributive justice used here in very general terms. For a much more detailed treatment, see Cozzens (2007).

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